Infrastructure and Platform as a Service (IaaS/PaaS)

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Introduction

- Infrastructure as a Service (IaaS) is a cloud computing delivery model that provides virtualized computing resources over the internet.
- Instead of purchasing and maintaining physical servers, storage, and networking equipment, users rent these resources on-demand from cloud service providers.
- IaaS forms the foundational layer of cloud computing (alongside PaaS and SaaS) and offers flexibility, scalability, and cost-effectiveness for businesses and developers.

Key characteristics include:

- **On-demand self-service** – Users can provision resources as needed without human intervention from the provider.
- **Scalability** – Capacity can be increased or decreased dynamically.
- **Pay-as-you-go pricing** – Customers pay only for the resources they consume.
- **Virtualization** – Resources are abstracted and delivered via virtual machines, containers, and software-defined networks.

- The evolution of IaaS is closely tied to advancements in virtualization, networking, and internet technologies.
- **Early days (Pre-2006)** – Organizations relied heavily on on-premises IT infrastructure. This approach involved high capital expenditure, long setup times, and limited scalability.
- **Rise of virtualization (2000–2006)** – Virtualization technologies (e.g., VMware, Xen, KVM) enabled multiple virtual machines to run on a single physical server, leading to more efficient utilization.
- **Launch of public IaaS services (2006 onwards)** – Amazon Web Services (AWS) launched EC2, followed by Google Cloud, Microsoft Azure, and others. These services allowed businesses to deploy and scale infrastructure globally in minutes.
- **Current trends** – Modern IaaS integrates automation, AI-driven resource management, hybrid cloud support, and enhanced security to meet diverse enterprise needs.

- A variety of academic and industry research has explored IaaS in terms of architecture, performance optimization, security, and integration.

Architecture & Resource Management

  - Studies such as Buyya et al. (2009) introduced cloud architecture frameworks like *Aneka* for resource allocation and workload scheduling in IaaS.
  - Research has focused on *orchestration* using tools like OpenStack, Kubernetes, and Terraform for automated provisioning.

Performance Optimization

- Works by Calheiros et al. (2011) proposed *CloudSim*, a simulation toolkit for modeling and evaluating IaaS performance.
- Dynamic load balancing, VM migration, and auto-scaling algorithms have been extensively studied to improve cost-efficiency.

Security & Compliance

- Research by Subashini and Kavitha (2011) highlighted vulnerabilities in IaaS, including VM isolation breaches, data leakage, and denial-of-service attacks.
- Solutions include encrypted storage, secure APIs, and compliance with standards like ISO 27001 and HIPAA.

Integration with Emerging Technologies

- Recent studies explore integrating IaaS with *edge computing* and *IoT* for low-latency applications.
- AI-based provisioning systems are being investigated for predictive scaling and energy-efficient cloud operations.

Virtual Provisioning in Virtualized Data Center

- VMP is capability to present a LUN to a computer system with extra capacity than what is physically allotted to the LUN on the storage array.

  It can be implemented by:

- Computer layer
- Storage layer
- Physical storage is allotted only when compute needs it and the provisioning decisions are not bound by presently accessible storage.
- The physical storage is allotted from a shared pool of physical capacity to the application “on-demand”.
- This gives more efficient storage utilization by minimizing the allotted amount, but unused physical storage

Need for Virtual Provisioning :

- Administrators typically allot storage space based on the anticipated growth of storage.
- This is because they want to minimize the management overhead and application downtime needed to add new storage afterwards.
- This results in the over-provisioning of the storage capacity which leads to greater costs, more power, cooling and floor space requirements and lower capacity utilization.
- Virtual Provisioning addresses these challenges by giving more efficient utilization of storage by minimizing the amount of allotted, but unused physical storage.

Advantages of Virtual Provisioning :

- Minimizes the operating and storage cost.
- Decreases downtime.
- Improves capacity utilization.
- Minimizes administrative overhead.

VIRTUAL MACHINES PROVISIONING AND MANAGEABILITY

- Virtual provisioning is a virtual storage network (VSAN)-based technology in which storage space is allocated on demand to devices.
- This process allows virtualized environments to control the allocation and management of physical disk storage connected with virtual machines (VM).
- Virtual Machine Management Service, also known as VMMS or vmms.exe, is the main module in Microsoft Windows operating system that controls all aspects of Hyper-V server virtualization.

VIRTUAL MACHINE MIGRATION SERVICES

- Virtual machine migration is the task of moving a virtual machine from one physical hardware environment to another.

VM Migration Type	Description
Cold Migration	Before migration, the virtual machine must be powered off, after doing this task. The old one should be deleted from source host. Moreover, the virtual machine need not to be on shared storage.
Warm Migration	Whenever transfer OS and any application, there is no need to suspend the source host. Basically it has high demand in public cloud.
Live Migration	It is the process of moving a running virtual machine without stopping the OS and other applications from source host to destination host.

TYPES OF VIRTUAL MACHINE MIGRATION TECHNIQUES

- Pre- Copy Migration
- Post-Copy Migration

PRE- COPY MIGRATION

- In this migration, the hypervisor copies all memory page from source machine to destination machine while the virtual machine is running.

It has two phases:

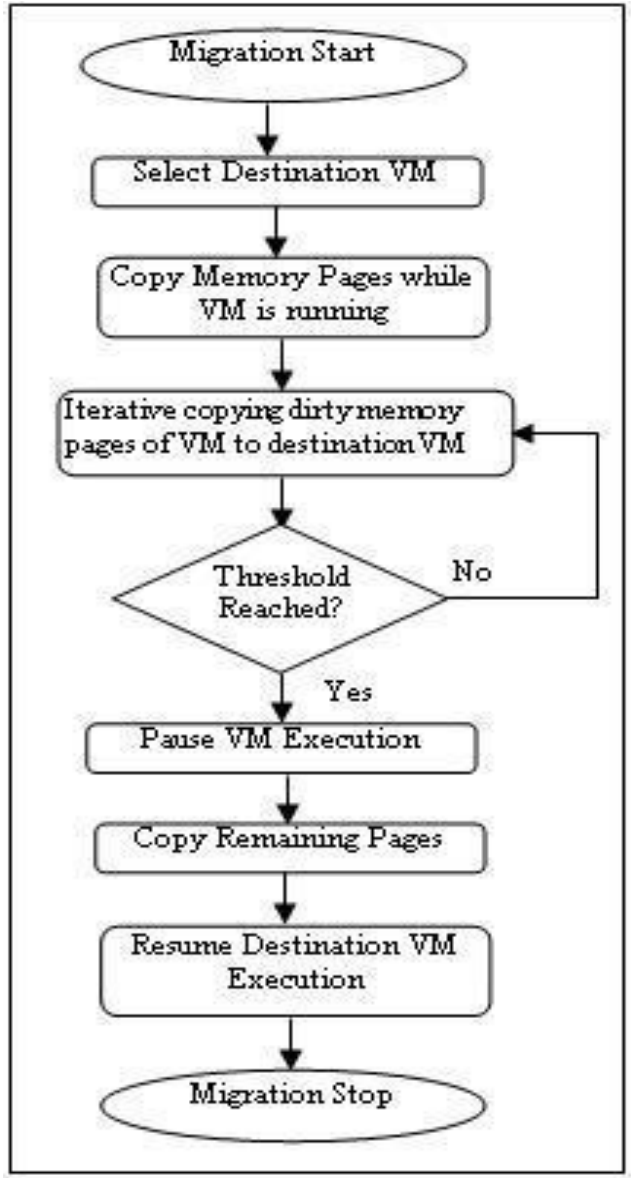
- Warm- up Phase and
- stop and copy phase.

WARM UP PHASE:

- During copying all memory pages from source to destination, some memory pages changed because of source machine CPU is active.
- All the changed memory paged known as dirty pages.
- All these dirty pages are required to recopy on destination machine; this phase is called as warm up phase.

STOP & COPY PHASE:

- Warm up phase is repeated until all the dirty pages recopied on destination machine.
- This time CPU of source machine is deactivated till all memory pages will transfer another machine.
- Ultimately at this time CPU of both source and destination is suspended, this is known as down time phase.
- This is the main thing that has to explore in migration for its optimization.

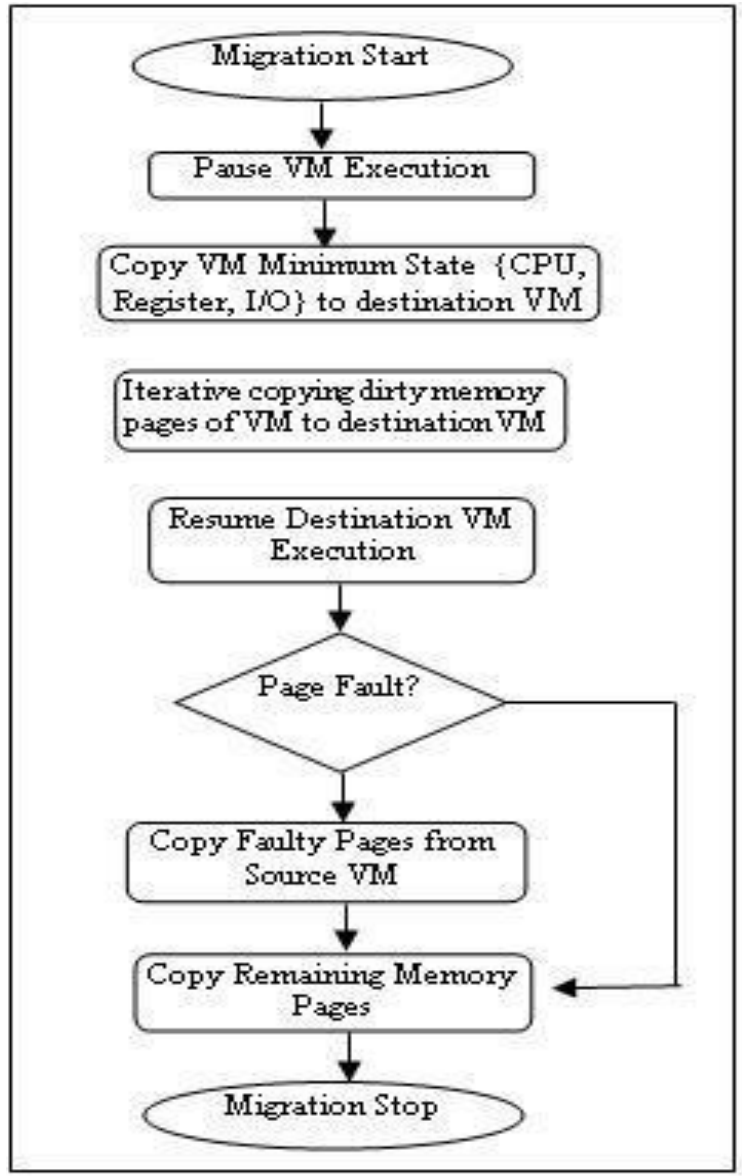


Pre- Copy Migration Technique

POST- COPY MIGRATION

- In this technique, VM at the source is suspended to start post copy VM migration.
- When VM is suspended, execution state of the VM (i.e. CPU state, registers, non-pageable memory) is transferred to the target.
- In parallel the sources actively send the remaining memory pages of the VM to the target. This process is known as pre-paging.

- At the target, if the VM tries to access a page that has not been transferred yet, it generates a page fault, also known as network faults.
- These faults are redirect to the source, which responds with the faulted pages.
- Due to this, the performance of applications is degrading with number of network faults.
- To overcome this, pre-paging scheme is used to push pages after the last fault by dynamically using page transmission order.



Post- Copy Migration Technique

Integration of Private and Public Cloud



HYBRID CLOUD

- Combination of both public and private cloud
- Shared security responsibility
- Helps maintain tighter controls over sensitive data and processes



PUBLIC CLOUD

- Offered by third-party providers
- Available to anyone over the public internet
- Scales quickly and convenient



PRIVATE CLOUD

- Offered to select users over the internet or a private internal network
- Provides greater security controls
- Requires traditional datacenter staffing and maintenance

Integrating public and private clouds hybrid cloud.

Private cloud: Provides control, security, and customization for sensitive data and applications.

Public cloud: Offers scalability, elasticity, and access to on-demand resources for flexible workloads.

Steps of integration of public and private clouds:

1. Identify Workloads and Data:

- Determine which workloads and data are suitable for the public cloud or the private cloud according to regulatory requirements, security concerns, or performance considerations.

2. Choose Integration Models:

- Cloud Bursting: dynamically scale workloads from the private cloud to the public cloud during periods of peak demand.

3. Implement Hybrid Cloud Management Platform:

- A hybrid cloud management platform (CMP) to orchestrate (the process of automating tasks to manage connections and operations of workloads on private and public clouds.).
- manage workloads across public and private clouds from a unified interface.
- Examples include VMware Cloud Foundation, Red Hat OpenShift, and Microsoft Azure Arc.

4. Optimize Workload Placement:

- Define policies and criteria for workload placement based on factors such as performance requirements, cost considerations, data sensitivity, and compliance regulations.

5. Ensure Security and Compliance:

- Implement consistent security policies, controls, and encryption mechanisms to protect data and workloads.

6. Disaster Recovery and Business Continuity:

- Implement a comprehensive disaster recovery (DR) and business continuity (BC) strategy that spans both public and private cloud environments.

7. Training and Documentation:

- Provide training to IT staff and users on managing and operating workloads in a hybrid cloud environment.

Challenges of Hybrid Cloud Integration:

Complexity: Managing two separate environments (private and public cloud) adds complexity.

Security Considerations: Data transfer between environments requires robust security measures to ensure data privacy and compliance.

Vendor Lock-in: Integration between different cloud providers can be challenging, potentially leading to vendor lock-in with a specific provider.

TECHNOLOGIES AND TOOLS FOR CLOUD COMPUTING

Virtualization: Virtualization technology in cloud computing involves the use of specialized programming techniques to create a virtual version of a computer resource to allow for cloud service providers to bring multiple operating systems and computer systems under one virtual machine to efficiently distribute an application among multiple users within one organization or numerous organizations in a cost-efficient and convenient manner.

- This cloud computing technique offers unparalleled hardware flexibility and significantly brings down IT maintenance and infrastructure costs, and is the first entry on the cloud computing technologies list.

- **Service-Oriented Architecture (SOA):** Service Oriented Architecture is the latest technology in cloud computing that refers to an online architecture that supports a collection of services.
- There is an established communication protocol between the components of various applications or services, and this communication occurs over a network to provide the user with certain services.

- **Grid Computing:** One of the other technologies used in cloud computing is grid computing, which allows for multiple computer resources to combine and interact to achieve a common objective.
- Grid computing is particularly beneficial when large-scale computing tasks are undertaken and are too burdensome for one individual computing resource to handle.
- Such a job is broken down into smaller components and distributed among the grid.
- The computing resources within the grid communicate with each other and work together to solve the problem.

- **Utility Computing:** Utility computing is one of the most common answers given to the question, ‘what is cloud-based technology?’
- Enabling technology for cloud computing, utility computing infrastructure is a software service provision business model that employs a pay-per-use model.
- Users can subscribe to preferred services and only pay for those services without undertaking a massive initial expenditure.
- This offers flexibility for users who wish to change or discontinue the usage of a cloud service.

RESOURCE PROVISIONING SERVICES

- Resource Provisioning means the selection, deployment, and run-time management of software (e.g., database server management systems, load balancers) and hardware resources (e.g., CPU, storage, and network) for ensuring guaranteed performance for applications.

METHODS OF RESOURCE PROVISIONIBG

Static Provisioning

- Predictable and Unchanging demands
- Advance preparation of resources
- Charges as flat free or monthly basis

Dynamic Provisioning

- Demand on application may change or vary
- Allocate more resources when needed and remove when it use

User-Self Provisioning

- Resources are purchased from web
- Create user account